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Roll No

ME-702(D)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Advance Machine Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What do you understand by Formative spur gear' in helical gears? 7
b) Explain the construction of involute profile. 7
2. a) A pair of parallel helical gears consists of 18 teeth pinion meshing with a 63 teeth gear. The helix angle is 23 degree while the normal pressure angle is 20 degree. Calculate axial pitch. 7
b) In which gear self-locking is possible? Give the condition for self-locking. 7
3. a) Find out the expression for maximum tension in belt. 7
b) Derive the expression for the length of crossed belt drive. 7
4. a) Write down the steps involve in designing Spur Gear. 7
b) Derive Lewis equation for bevel gears. 7
5. a) What are the modes of failure for helical gears. 7
b) Write down the General design considerations in IC engines. 7

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6. a) Write down the general steps involved in design of bevel gear. 7
b) What are the forces involved in bevel gear while in operation explain. 7
7. a) Write short notes on knuckle and cotter joints. 7
b) Design a connecting rod for a petrol engine from the following data: Diameter of piston = 110 mm; Mass of reciprocating parts = 2 kg; Length of connecting rod = 325 mm; Stroke = 150 mm; Speed = 1500 rpm with possible over speed of 1850 rpm; Compression ratio = 4:1; Factor of safety = 4; Maximum explosion pressure = 5.5 MPa Select suitable material and permissible stresses for its. 7
8. a) Design a plain carbon steel centre crankshaft for a single acting four stroke, single cylinder engine for the following data: Piston diameter = 250 mm; Stroke = 400 mm; Maximum combustion pressure = 2.5 MPa; Weight of the flywheel = 5 kg; Total belt pull = 100 N; Length of connecting rod = 950 mm. The flywheel is used as a pulley. When the crank has turned through 300 from top dead centre, the pressure on the piston is 1 MPa and the torque on the crank is maximum. Any other data required for the design may be assumed. 7
b) It is required to design a pair of bevel gears, which are mounted on shafts intersecting at right angles. The pinion receives 20 kW power through its shaft and rotates at 720 rpm. The number of teeth on pinion and gear are 30 and 45 respectively. The pressure angle is 20° full depth teeth form the gears are made of plain carbon steel with permissible bending stress as 200 MPa. The gears are case hardened and the surface hardness is 300 BHN. Take service factor = 1.25. 7
